

Machine Learning Lab -1	
Course Code: ISL56	Credits: 0:0:1
Pre – requisites: Python	Contact Hours: 14P/15P
Course Coordinator: Dr. Shruti G	

Course Content

PART -A

1. Exploring and Visualizing Sales Data. Import a sample sales dataset into Tableau. Create a dashboard that includes the following visualizations:
 1. A bar chart showing total sales by product category.
 2. A line chart showing monthly sales trends over the past year.
 3. A geographic map showing sales distribution by region.
 4. Use a dual-axis chart or side-by-side bar charts to compare sales and return rates across regions.
2. Using a dataset on customer purchases, create a dashboard with the following advanced visualizations:
 1. A scatter plot showing the relationship between customer age and total spending.
 2. A heat map illustrating product sales intensity by store location.
 3. A dual-axis chart comparing sales and profit margins over time.
 4. Create a bar chart or treemap showing total sales revenue by product and region.
3. Analysing Sales Performance: Create various visualizations to analyze the sales performance.
 1. Create a line chart to show sales trends over time.
 2. Create a bar chart to compare sales across different regions.
 3. Use a pie chart to display the distribution of sales among different product categories.
 4. Create a dashboard combining these visualizations to provide an overall view of sales performance.

PART – B

1. Dataset (<https://archive.ics.uci.edu/ml/datasets/wine+quality>)
 - (a) Analyze and visualize the dataset to understand the relationship between physicochemical features and wine quality.
 - (b) Build a regression model using Linear Regression and Ridge Regression.
 - (c) Evaluate and interpret the model performance using R^2 , RMSE, and MAE metrics.
2. Dataset (<https://archive.ics.uci.edu/ml/datasets/heart+disease>)
 - (a) Analyze and visualize the dataset to understand patterns and relationships among clinical features and heart disease outcomes.
 - (b) Build a classification model using Logistic Regression and apply a variant approach using Multinomial Logistic Regression.
 - (c) Evaluate and interpret the model performance using Accuracy, Precision, Recall, F1-score, and ROC-AUC metrics.
3. Dataset <https://archive.ics.uci.edu/ml/datasets/parkinsons>

- (a) Analyze and visualize the dataset to explore correlations and relationships among features related to Parkinson's disease.
- (b) Build a classification model using K-Nearest Neighbors (KNN) and apply a variant approach using Weighted KNN. Use the Elbow Method to determine the optimal value of k .
- (c) Evaluate and interpret the model performance using Accuracy and Confusion Matrix metrics.

4. Dataset ([https://archive.ics.uci.edu/ml/datasets/breast+cancer+wisconsin+\(diagnostic\)](https://archive.ics.uci.edu/ml/datasets/breast+cancer+wisconsin+(diagnostic)))

- (a) Analyze and visualize the dataset to explore feature correlations and patterns among diagnostic attributes.
- (b) Apply Principal Component Analysis (PCA) for dimensionality reduction and build a classification model using Support Vector Machine (SVM) with a variant approach using Kernel SVM (RBF).
- (c) Evaluate and interpret the model performance using Accuracy and ROC-AUC metrics.

5. Dataset

(<https://archive.ics.uci.edu/ml/datasets/human+activity+recognition+using+smartphones>)

- (a) Analyze and visualize the dataset to explore feature distributions and relationships among activity-related attributes.
- (b) Build a classification model using Random Forest and apply a variant approach using Extra Trees Classifier.
- (c) Evaluate and interpret the model performance using Accuracy and Macro-F1 Score metrics.

6. Dataset (<https://archive.ics.uci.edu/ml/datasets/spambase>)

- (a) Analyze and visualize the dataset to understand the distribution and characteristics of spam-related features. Analyze the UCI Spambase dataset to understand the distribution and characteristics of spam-related features such as word frequency, character frequency, and capital run length. Visualize the dataset using appropriate plots to study the patterns of spam and non-spam emails.
- (b) Build a classification model using Naïve Bayes and apply a variant approach using Gaussian Naïve Bayes.
- (c) Evaluate and interpret the model performance using Accuracy, Precision, Recall, and F1-score metrics.

7. Dataset (<https://archive.ics.uci.edu/ml/datasets/default+of+credit+card+clients>)

- (a) Analyze the dataset to study class distribution and visualize the imbalance between defaulters and non-defaulters.
- (b) Build a baseline Random Forest classification model for default prediction and apply data balancing techniques such as oversampling, undersampling, and SMOTE.
- (c) Evaluate the models using Precision, Recall, F1-score, ROC-AUC, and PR-AUC. Implement Gradient Boosting as a variant algorithm and compare the performance of models with and without data balancing.

8. Dataset <https://archive.ics.uci.edu/ml/datasets/bank+marketing>

- (a) Analyze the Bank Marketing dataset and visualize the distribution of customers subscribing to term deposits. Use appropriate plots to study class distribution and the behavior of numerical features.
- (b) Develop a Bayesian classification model to predict whether a customer will subscribe to a term deposit. Explain the working principle of the Bayesian classifier.

(c) Evaluate the model using Accuracy and ROC-AUC metrics.
Implement Logistic Regression as a variant algorithm and compare its performance with the Bayesian classifier. Interpret the results and justify the better-performing model.

9. Dataset

<https://archive.ics.uci.edu/ml/datasets/optical+recognition+of+handwritten+digits>

- (a) Analyze the Optical Recognition of Handwritten Digits dataset and visualize sample digit images and feature patterns using appropriate plots.
- (b) Build a classification model using the Multilayer Perceptron (MLP) method for handwritten digit recognition. Implement a variant MLP with different hidden layer sizes and discuss the effect of network architecture on performance.
- (c) Evaluate the models using Accuracy and Confusion Matrix measures, and compare the performance of the baseline and variant MLP models.

Text Book:

- 13. S Sridhar, M Vijayalakshmi, “Machine Learning”, OXFORD University Press 2021, First Edition.
- 14. T. M. Mitchell, “Machine Learning”, McGraw Hill, 1997.

Reference Books:

- 1. M N Murty and Ananthanarayana V S, “Machine Learning: Theory and Practice”, Universities Press (India) Pvt. Limited, 2024.
- 2. Stephen Marsland, “Machine Learning - An Algorithmic Perspective”, Second Edition, CRC Press -Taylor and Francis Group, 2015

Self-Learning (Programming Assignment/ Case Study/ Project Based Learning/ MOOC based learning / GATE oriented aptitude test / Simulation):

Course Outcomes(COs):

At the end of the course, the student will be able to:

Course outcomes (Cos):

At the end of the course the student will be able to:

- 1. Understand and implement various machine learning procedures algorithms using appropriate datasets.(PO: 1, 2, 3, 4, 5, PSO: 1, 3)
- 2. Design and develop Java/Python programs for supervised, unsupervised, and ensemble learning methods. (PO: 1, 2, 3, 4, 5, PSO: 1, 3)
- 3. Apply machine learning techniques to analyze and solve real-world problems. (PO: 1, 2, 3, 4, 5, PSO: 1, 3)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment tool	Marks	Course outcomes attained
Internal Test	20	CO1, CO2, CO3
Other Components:		
Weekly Evaluation- Lab Record	20	CO1, CO2, CO3
Quiz	10	CO1, CO2, CO3
The Final CIE out of 50 Marks = Marks Scored in Lab Test + Marks of Lab Record + Marks Scored in Quiz		

Conduction, Calculation, Record and Viva	30	CO1, CO2, CO3
Semester-End Examination (SEE)	50	CO1, CO2, CO3